Sara Petiton
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Professional Experience

Oct. 2022 - May 2026 **PhD contract** | NeuroSpin — CEA Paris-Saclay | Gif-sur-Yvette, France
PhD in Machine and Deep Learning applied to neuroimaging and biomarker discovery.

Oct. 2021 - Jul. 2022 **Software Engineer** | Intel Corporation | Meudon, France
Developed Deep Learning models to predict CPU performance.

Apr. 2021 - Oct. 2021 **Software Engineer Intern** | Intel Corporation | Meudon, France
Six-month internship focused on preliminary modeling for CPU performance prediction.

Jul - Aug. 2017 **Data Science Intern** | Global Footprint Network | Oakland, CA
Analyzed databases (from the UN, International Energy Agency, and more) using SQL and R to determine the ecological footprint of different countries.

Publications

- Dufumier B., Gori P., **Petiton S.** et al., "Exploring the potential of representation and transfer learning for anatomical neuroimaging: Application to psychiatry", *NeuroImage*, Vol. 296, 2024, doi: 10.1016/j.neuroimage.2024.120665
- **Petiton S.**, Grigis A., Dufumier B., Duchesnay E., "How and Why Does Deep Ensemble Coupled with Transfer Learning Increase Performance in Bipolar Disorder and Schizophrenia Classification?", *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, pp. 1-5, doi: 10.1109/ISBI56570.2024.10635777
- **Petiton S.**, Grigis A., Dufumier B., et al. "Surface and voxel-based brain morphometric representations are synergistic for individual-level prediction of bipolar disorder", *Hal preprint*, 2026, HAL Id: hal-05571683.
- **Petiton S.**, Bourgin J., Grigis A., et al. "Hippocampal and amygdalar atrophy predicts lithium response in bipolar disorder: machine learning evidence from a prospective multi-site cohort", *Hal preprint*, 2026 (work part of the R-Link European initiative), HAL Id: inserm-05588008.
- **Petiton S.**, Grigis A., Vock R., Duchesnay E. "Improving clinical interpretability of linear neuroimaging models through feature whitening", *arXiv preprint*, 2026, arXiv:2604.20675

Research work

- **High-Performance Computing**: Led large-scale DL/ML experiments on **Jean Zay (IDRIS)** and NeuroSpin clusters using multi-site datasets (UK Biobank, OpenBHB, ABIDE 1+2, BSNIP, and others).
- **Clinical ML**: Developed predictive models for Bipolar Disorder classification and lithium response prediction (part of the **RLink European Project**).
- **Journal Club**: Participated in and co-led the deep learning journal club at NeuroSpin during my PhD.
- **Scientific Communication**: Presented research at **OHBM 2023** (*Organization for Human Brain Mapping,* Montreal), **IABM 2024** (*French Conference on AI in Biomedical Imaging,* Grenoble), and **ISBI 2024** (*International Symposium on Biomedical Imaging,* Athens).
- **Software**: Developed an interactive web app mapping normative trajectories of regional brain volumes across age and sex (Neuromorphometrics atlas). https://sarampetiton.github.io/brain-normative-atlas-explorer/

Research-focused summer schools

Jul. 2026 **AI for Science** | **Hellenic Institute of Advanced Studies** and **Google DeepMind** | Athens, Greece
Selected for a two-day program focused on ML applications in scientific discovery. Topics included topological deep learning, physics-enhanced neural networks, and AlphaGenome, among others. Additionally presented some of my PhD research.

Jul. 2025 **Computational Neuroscience** | **Cold Spring Harbor Asia** (CSHA) | Suzhou, China
Selected for an intensive month-long program focused on neural mechanisms of cognitive functions. Covered topics from single-neuron dynamics to AI in psychiatric disorders. Participated in advanced programming practicals.

Teaching experience

Spring semester 2023 Teaching Assistant (Java/OOP) | Paris-Saclay Computer Science IUT
Taught and mentored first-year students in Java-based Object-Oriented Programming concepts.

Education

Oct. 2022 - May 2026 **PhD** | University Paris-Saclay and NeuroSpin (CEA), BAOBAB Research Unit, GAIA lab

Title: ***Machine and Deep Learning methods for identifying neuroanatomical biomarkers of bipolar disorder.***

Supervised by Dr. Edouard DUCHESNAY and Dr. Antoine GRIGIS. Manuscript available at HAL Id: tel-05658678

2019 - 2021 Université Paris Cité (Paris V – Descartes) | Paris, France
Master of Applied Mathematics, Specialty: Mathematics, Modeling, Learning

2015 - 2019 Sorbonne Université (Paris VI – UPMC) | Paris, France
Bachelor of Mathematics, Minor in Computer Science

Summer 2019 University of California, Berkeley | Berkeley, CA, USA | Summer Semester
Coursework in Multimedia, Investigative Reporting, and Stress/Coping (8 units)

Summer 2016 University of California, Berkeley | Berkeley, CA, USA | Summer Semester
Coursework in Statistics and Economic Journalism (7 units)

Jun. 2015 École Jeannine Manuel | Scientific Baccalauréat, OIB American Section (Honors, Mathematics Specialty) & NEASC-Accredited HS Diploma

Languages

French Native Speaker

English Fluent (SAT, IGCSE/Cambridge, American High School Diploma)

Spanish Proficient in speaking, reading, and writing

Technical skills

Languages: Python, Java, SQL, R, Bash, C, Matlab

AI & Data Science: Pytorch, Scikit-learn, Pandas, Nibabel and Nilearn (for neuroscience data)

Undergraduate Projects: Extreme Value Theory (Statistics, supervised by J. El Methni, Paris Descartes), PageRank & Markov Chains (Python), Cryptology & Error Correction Codes (supervised by R. Barbulescu, Sorbonne Université), Java Car Race Simulator (OOP)

Interests & Activities

Violinist: classical training since childhood as part of a CHAM program (a selective specialized curriculum balancing intensive music theory/practice with academics); participated in multiple orchestras. Alumna of the Regional Conservatory of Boulogne-Billancourt (8 years). Participated in a masterclass led by Kent Nagano (Paris, 2014). Also enjoy running, yoga, cooking, and reading.